

observation

Step 1 – Customer table cleaning

Replacing inconsistent data, incorrect datatype, using proper function for Customer name, uppercase country, checking for duplicate values and trim() to avoid unwanted space.

Table.TransformColumns(#'Filtered Rows',{{"Country", Text.Upper, type text}})

Customer_ID	Name	Age	Gender	City	State	Country
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA
CUST1004	Caroline Collins	64	Female	Nicholashire	California	USA
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA

Step 2: filling null values in loyalty level as per majority level. Platinum was majority and I fill Platinum in null places

=Platinum

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Platinum
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholashire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Platinum
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze

Step 3: Product dim cleaning

Replacing inconsistent data, incorrect datatype, using proper function for product name, checking for duplicate values and trim() to avoid unwanted space.

Table1 - Power Query Editor

Query Settings

Table1

Properties

Applied Steps

Source

Changed Type

Replaced product id P-3 to PRL

Replaced P-16 to PRO016

Capitalized Each Word

Trimmed Text

Capitalized Each Word1

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PRO00	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	
PRO01	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	
PRO02	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	null	
PRO03	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	
PRO04	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	
PRO05	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza And Gonzalez	261	
PRO06	Herbal Face Moisturizer	Beauty	Skincare	Davidson Plc	312.8	
PRO07	Led Table Lamp	Home	Lighting	Harrell Llc	218.86	
PRO08	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	
PRO09	Adjustable Dumbbell Set	Sports	Fitness	Hill Plc	495.72	
PRO010	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	
PRO011	Hd Smart Tv Stick	Electronics	Accessories	Gibson, Hobbs And Weaver	449.4	
PRO012	Hair Strengthening Shampoo	Beauty	Haircare	Rice Llc	115.82	
PRO013	Usb-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	
PRO014	Decorative Wall Mirror	Home	Decor	Luna, Anderson And Bowman	48.01	
PRO015	4K Home Projector	Electronics	Appliances	Thomas Llc	465.34	
PRO016	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	
PRO017	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy And Wilson	144.53	
PRO018	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	null	
PRO019	Knee Support Brace	Sports	Equipment	May-Reese	21.34	
PRO020	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	
PRO021	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	
PRO022	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda And Rodriguez	484.62	
PRO023	Men's Formal Shirt	Clothing	Tops	Moon, Mora And Santos	null	
PRO024	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	
PRO025	Women's Scarf Set	Clothing	Accessories	Brown And Sons	174.02	
PRO026	Wireless Bluetooth Speaker	Electronics	Audio	Reed-Schmidt	322.41	

Step 4 : filling Cost null values with average as per category wise.

Cleaned_Ecommerce_Sales_Dataset

Product_ID

Product_Name

Category

Sub_Category

Brand

Cost

Stock

average for clothing 265.7327

avg for home 236.5637

avg for beauty 250.4162

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PRO06	Herbal Face Moisturizer	Beauty	Skincare	Davidson Plc	312.8	350
PRO012	Hair Strengthening Shampoo	Beauty	Haircare	Rice Llc	115.82	460
PRO016	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PRO032	Vitamin C Face Serum	Beauty	Skincare	Griffin Ltd	283.83	244
PRO047	Makeup Brush Set	Beauty	Makeup	Cook Inc	250.42	431
PRO049	Men's Roll-On Perfume	Beauty	Fragrance	Sullivan And Sons	43.29	267
PRO059	Men's Hair Styling Gel	Beauty	Haircare	Cline-Mccarthy	17.31	399
PRO060	Luxury Perfume Spray	Beauty	Fragrance	Gibson, Green And Lopez	482.75	444
PRO063	Organic Sunscreen Lotion	Beauty	Skincare	Anderson-Riley	266.86	458
PRO064	Matte Lipstick Set	Beauty	Makeup	Zhang Inc	299.11	118
PRO066	Women's Hair Dryer Brush	Beauty	Haircare	Torres, Spencer And Potter	340.39	177
PRO071	Hair Repair Oil Treatment	Beauty	Haircare	Hopkins Ltd	214.32	383
PRO083	Hydrating Face Cleanser	Beauty	Skincare	Butler Group	250.42	159
PRO089	Face Foundation Kit	Beauty	Makeup	Taylor-Elliott	183.04	
PRO094	Vitamin E Body Lotion	Beauty	Skincare	Hoffman-Williams	358.47	

Same for stock null values, I filled as per category.

AutoSave Off Cleaned_Ecommerce_Sales_Dataset Search Upgrade your plan

File Home Insert Draw Page Layout Formulas Data Review View Help

Clipboard Font Alignment Number Styles Cells Editing Add-ins

K19 fx

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men'S Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women'S Casual Hoodie	Clothing	Outerwear	Sims Inc	265.73	352
PROD3	Men'S Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women'S Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men'S Leather Belt	Clothing	Accessories	Baldwin, Garza And Gonzalez	261	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson Plc	312.8	350
PROD7	Led Table Lamp	Home	Lighting	Harrell Llc	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill Plc	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	Hd Smart Tv Stick	Electronics	Accessories	Gibson, Hobbs And Weaver	449.4	275
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice Llc	115.82	460
PROD13	Usb-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson And Bowman	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas Llc	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy And Wilson	144.53	367
PROD18	Women'S Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	265.73	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	282
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda And Rodriguez	484.62	432
PROD23	Men'S Formal Shirt	Clothing	Tops	Moon, Mora And Santos	265.73	360

Sales Fact Cleaned_Customer_dim Cleaned_product_dim cleaned_Store_dim Cleaned_sales_fact

Ready 29°C Partly sunny Search 12:48 05-10-2026

Step 5: Store Dim cleaning

Basic cleaning using trim , proper and datatype checking. No null and duplicated present .

Table5 - Power Query Editor

File

Home

Transform

Add Column

View

Close & Load

Refresh

Advanced Editor

Properties

Manage

Query

Choose Columns

Remove Columns

Manage Columns

Keep Rows

Remove Rows

Reduce Rows

Sort

Split Column

Group By

Data Type: Text

Use First Row as Headers

Replace Values

Transform

Merge Queries

Append Queries

Combine Files

Combine

Manage Parameters

Parameters

Data source settings

Data Sources

New Source

Recent Sources

Enter Data

New Query

Queries [3]

Table1

Table3

Table5

Table.TransformColumns(#"Capitalized Each Word",{{"Store_Type", Text.Trim, type text}, {"City", Text.Trim, type text}, {"Region",

Store_ID

Store_Name

Region

City

Store_Type

1

Rivera-Warner

East

Washingtonfurt

Flagship

2

STORE1

Martinez, Adkins And Johnson

North

Danielmouth

Flagship

3

STORE2

Black-Horn

South

North Nancyview

Outlet

4

STORE3

Johnson-Wright

North

Hoodberg

Outlet

5

STORE4

Alexander Group

South

Port Sharon

Online

6

STORE5

Gray Ltd

North

New Michaelchester

Outlet

7

STORE6

Benson-Elliott

East

Port Josephberg

Online

8

STORE7

Martinez-Mullins

South

Williamsfurt

Online

9

STORE8

Bean, Jones And Park

West

Thompsonborough

Online

10

STORE9

Moore, Padilla And Butler

South

East Johnstad

Flagship

11

STORE10

Hill, Wagner And Ballard

East

East Davidville

Online

12

STORE11

Aguiar-Keller

South

Lake Michael

Online

13

STORE12

Thomas, Bridges And Harris

North

Sabrinatown

Online

14

STORE13

Gardner, Humphrey And Contreras

East

Johnsonville

Flagship

15

STORE14

McDonald Inc

South

Christopherland

Outlet

16

STORE15

Shelton Plc

North

East Jenniferstad

Flagship

17

STORE16

Frederick-Wright

West

New Jimmymouth

Flagship

18

STORE17

Holloway Ltd

North

Bensonhaven

Flagship

19

STORE18

Guerrero-Schmitt

East

East Stevechester

Flagship

20

STORE19

Mann-Austin

East

Vasquezborough

Online

Query Settings

PROPERTIES

Name

Table5

All Properties

APPLIED STEPS

Source

Changed Type

Capitalized Each Word

Trimmed Text

5 COLUMNS, 20 ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 01:33

23°C

Clear

Search

Step 6: Sales Fact Cleaning.

Basic cleaning in power query using trim , proper and datatype checking. No null and duplicated present .

Table7 - Power Query Editor

Formula Bar: `= Table.TransformColumnTypes(#Filtered Rows,{{"Unit_Price", type number}})`

	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
1	STORE3	3	218.04	0.09	Credit Card	198.41
2	STORE17	3	104.47	0.27	PayPal	228.78
3	STORE8	4	914.05	0.25	Credit Card	2742.
4	STORE8	3	921.51	0.16	COD	774.06
5	STORE5	3	745.3	0.18	Credit Card	1833.4
6	STORE7	3	147.01	0.19	COD	357.23
7	STORE4	2	785.31	0.15	PayPal	1385.0
8	STORE8	4	342.39	0.23	PayPal	1054.56
9	STORE5	4	448.25	0.07	PayPal	1250.61
10	STORE0	2	873.17	0.23	PayPal	1344.68
11	STORE14	1	278.63	0.07	Credit Card	259.12
12	STORE1	4	286.92	0.17	Credit Card	952.57
13	STORE4	4	644.58	0.01	Credit Card	2552.53
14	STORE16	2	316.9	0.3	PayPal	443.
15	STORE15	4	403.47	0.29	PayPal	1145.85
16	STORE9	4	871.34	0.04	COD	2509.45
17	STORE19	2	451.98	0.25	Credit Card	677.
18	STORE4	2	502.69	0.23	Credit Card	774.14
19	STORE4	2	559.23	0.25	COD	838.8
20	STORE13	4	443.	0.04	COD	1782.45
21	STORE17	3	986.04	0.27	COD	2159.42
22	STORE18	1	589.56	0.22	Credit Card	459.85
23	STORE10	3	979.42	0.03	PayPal	2850.11
24	STORE8	1	719.28	0.08	Credit Card	661.73
25	STORE6	3	77.6	0.07	Credit Card	216.5
26	STORE3	4	852.71	0.14	PayPal	2933.32
27	STORE8	4	156.16	0.14	Credit Card	599.65

Filled null vales in quantity as per store category.

Cleaned_Ecommerce_Sales_Dataset

Formula Bar: `=J45/(G45*(1-H45))`

	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
10	SALE8	04-05-2025	CUST1187	PROD99	STORE5	3	448.25	0.07	PayPal	1250.6175
17	SALE15	10-12-2024	CUST11435	PROD36	STORE9	3	871.34	0.04	COD	2509.4592
31	SALE29	11-01-2024	CUST1348	PROD99	STORE3	4	451.67	0.15	Credit Card	1535.878
36	SALE34	16-10-2023	CUST1344	PROD79	STORE1	3	873.22	0.18	Credit Card	2148.1212
45	SALE43	20-01-2025	CUST1084	PROD33	STORE17	3	863.11	0.28	PayPal	1864.3176

Hence one row has empty unit price and discount. So I filled discount value by taking average as per store and filled only 1 null values in discount column and later Filled null values in unit price using formula.

Unitprice= total amount/[quantity*(1-discount)]

Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
18-03-2024	CUST1239	PROD060	STORE13	4	464.18	0.04	COD	1782.4512
25-08-2025	CUST1148	PROD089	STORE6	3	190.6130588	0.15	Credit Card	486.0633
20-09-2024	CUST1406	PROD090	STORE18	3	111.79	0.06	Credit Card	315.2478

Later filled discount values using formula

Discount = 1 – (Total_Amount ÷ (Quantity × Unit_Price))

Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
03-01-2024	CUST1213	PROD098	STORE8	4	156.16	0.04	Credit Card	599.6544
07-12-2024	CUST1404	PROD15	STORE7	4	733.42	0.13	PayPal	2552.3016
26-04-2025	CUST1248	PROD35	STORE2	2	355.3	0.17	PayPal	589.798

Filled total amount null values using formula

Discount = 1 – (Total_Amount ÷ (Quantity × Unit_Price))

	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
66	14-09-2025	CUST1255	PROD53	STORE10	3	480.06	0.01	Credit Card	1425.7782
72	11-07-2025	CUST1058	PROD60	STORE5	3	92.24	0.28	COD	199.2384
86	08-12-2023	CUST1423	PROD46	STORE17	1	773.57	0.05	COD	734.8915
96	26-08-2024	CUST1298	PROD97	STORE19	2	569.36	0.01	PayPal	1127.3328

Step 7: using Vlookup and Xlookup

Using Vlookup I bought the category in salesfact table.

	Sales_ID	Order_Date	Customer_ID	Product_ID	Category	Store_ID	Store_name	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
1	SALE0	07-07-2024	CUST1409	PROD97	Sports	STORE3	Johnson-Wright	1	218.04	0.09	Credit Card	198.4164
2	SALE1	25-12-2023	CUST1467	PROD74	Electronics	STORE17	Holloway Ltd	3	104.47	0.27	PayPal	228.7893
3	SALE2	03-06-2024	CUST1056	PROD49	Beauty	STORE8	Bean, Jones And Park	4	914.05	0.25	Credit Card	2742.15
4	SALE3	29-11-2023	CUST1346	PROD76	Electronics	STORE8	Bean, Jones And Park	1	921.51	0.16	COD	774.0684
5	SALE4	06-01-2024	CUST1493	PROD59	Beauty	STORE5	Gray Ltd	3	745.3	0.18	Credit Card	1833.438
6	SALE5	26-07-2024	CUST1116	PROD15	Electronics	STORE7	Martinez-Mullins	3	147.01	0.19	COD	357.2343
7	SALE6	12-07-2025	CUST1016	PROD56	Home	STORE4	Alexander Group	2	785.31	0.15	PayPal	1335.027
8	SALE7	18-11-2023	CUST1152	PROD70	Home	STORE8	Bean, Jones And Park	4	342.39	0.23	PayPal	1054.5612
9	SALE8	04-05-2025	CUST1187	PROD99	Home	STORE5	Gray Ltd	3	448.25	0.07	PayPal	1250.6175
10	SALE9	31-07-2025	CUST1464	PROD33	Home	STORE0	Rivera-Warner	2	873.17	0.23	PayPal	1344.6818
11	SALE10	13-02-2024	CUST1065	PROD24	Sports	STORE14	Mcdonald Inc	1	278.63	0.07	Credit Card	259.1259
12	SALE11	24-05-2025	CUST1237	PROD39	Electronics	STORE1	Martinez, Adkins And Jc	4	286.92	0.17	Credit Card	952.5744
13	SALE12	24-08-2024	CUST1237	PROD88	Sports	STORE4	Alexander Group	4	644.58	0.01	Credit Card	2552.5368
14	SALE13	06-08-2025	CUST1443	PROD34	Electronics	STORE16	Frederick-Wright	2	316.9	0.3	PayPal	443.66
15	SALE14	21-06-2025	CUST1348	PROD24	Sports	STORE15	Shelton Plc	4	403.47	0.29	PayPal	1145.8548
16	SALE15	10-12-2024	CUST1435	PROD36	Clothing	STORE9	Moore, Padilla And But	3	871.34	0.04	COD	2509.4592
17	SALE16	13-09-2025	CUST1276	PROD65	Sports	STORE19	Mann-Austin	2	451.98	0.25	Credit Card	677.97
18	SALE17	11-12-2024	CUST1079	PROD27	Home	STORE4	Alexander Group	2	502.69	0.23	Credit Card	774.1426
19	SALE18	13-08-2024	CUST1094	PROD39	Electronics	STORE4	Alexander Group	2	559.23	0.25	COD	838.845
20	SALE19	18-03-2024	CUST1239	PROD60	Beauty	STORE13	Gardner, Humphrey And	4	464.18	0.04	COD	1782.4512
21	SALE20	05-07-2025	CUST1144	PROD72	Sports	STORE17	Holloway Ltd	3	986.04	0.27	COD	2159.4276
22	SALE21	04-06-2024	CUST1396	PROD43	Electronics	STORE18	Guerrero-Schmitt	1	589.56	0.22	Credit Card	459.8568
23	SALE22	12-04-2024	CUST1476	PROD69	Clothing	STORE10	Hill, Wagner And Ballan	3	979.42	0.03	PayPal	2850.1122
24	SALE23	23-06-2024	CUST1368	PROD12	Beauty	STORE8	Bean, Jones And Park	1	719.28	0.08	Credit Card	661.7376
25	SALE24	08-09-2025	CUST1186	PROD67	Electronics	STORE6	Benson-Elliott	3	77.6	0.07	Credit Card	216.504
26	SALE25	17-09-2024	CUST1381	PROD5	Clothing	STORE3	Johnson-Wright	4	852.71	0.14	PayPal	2933.3224

Using Xlookup I bought store name in sales fact table.

AutoSave OFF Cleaned_Ecommerce_Sales_Dataset Search Upgrade your plan

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Clipboard Font Alignment Number Styles Cells Editing Add-ins

G4 XLOOKUP(F4,Cleaned_Store_dim!A:A,Cleaned_Store_dim!B:B,"Not Found")

Sales_ID	Order_Date	Customer_ID	Product_ID	Category	Store_ID	Store_name	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	07-07-2024	CUST1409	PROD97	Sports	STORE3	Johnson-Wright	1	218.04	0.09	Credit Card	198.4164
SALE1	25-12-2023	CUST1467	PROD74	Electronics	STORE17	Holloway Ltd	3	104.47	0.27	PayPal	228.7893
SALE2	03-06-2024	CUST1056	PROD49	Beauty	STORE8	Bean, Jones And Park	4	914.05	0.25	Credit Card	2742.15
SALE3	29-11-2023	CUST1346	PROD76	Electronics	STORE8	Bean, Jones And Park	1	921.51	0.16	COD	774.0684
SALE4	06-01-2024	CUST1493	PROD59	Beauty	STORE5	Gray Ltd	3	745.3	0.18	Credit Card	1833.438
SALE5	26-07-2024	CUST1116	PROD15	Electronics	STORE7	Martinez-Mullins	3	147.01	0.19	COD	357.2343
SALE6	12-07-2025	CUST1016	PROD56	Home	STORE4	Alexander Group	2	785.31	0.15	PayPal	1335.027
SALE7	18-11-2023	CUST1152	PROD70	Home	STORE8	Bean, Jones And Park	4	342.39	0.23	PayPal	1054.5612
SALE8	04-05-2025	CUST1187	PROD99	Home	STORE5	Gray Ltd	3	448.25	0.07	PayPal	1250.6175
SALE9	31-07-2025	CUST1464	PROD33	Home	STORE0	Rivera-Warner	2	873.17	0.23	PayPal	1344.6818
SALE10	13-02-2024	CUST1065	PROD24	Sports	STORE14	McDonald Inc	1	278.63	0.07	Credit Card	259.1259
SALE11	24-05-2025	CUST1320	PROD39	Electronics	STORE1	Martinez, Adkins And Jc	4	286.92	0.17	Credit Card	952.5744
SALE12	24-08-2024	CUST1237	PROD88	Sports	STORE4	Alexander Group	4	644.58	0.01	Credit Card	2552.5368
SALE13	06-08-2025	CUST1443	PROD34	Electronics	STORE16	Frederick-Wright	2	316.9	0.3	PayPal	443.66
SALE14	21-06-2025	CUST1348	PROD24	Sports	STORE15	Shelton Plc	4	403.47	0.29	PayPal	1145.8548
SALE15	10-12-2024	CUST1435	PROD36	Clothing	STORE9	Moore, Padilla And But	3	871.34	0.04	COD	2509.4592
SALE16	13-09-2025	CUST1276	PROD65	Sports	STORE19	Mann-Austin	2	451.98	0.25	Credit Card	677.97
SALE17	11-12-2024	CUST1079	PROD27	Home	STORE4	Alexander Group	2	502.69	0.23	Credit Card	774.1426
SALE18	13-08-2024	CUST1094	PROD39	Electronics	STORE4	Alexander Group	2	559.23	0.25	COD	838.845
SALE19	18-03-2024	CUST1239	PROD60	Beauty	STORE13	Gardner, Humphrey Ani	4	464.18	0.04	COD	1782.4512
SALE20	05-07-2025	CUST1144	PROD72	Sports	STORE17	Holloway Ltd	3	986.04	0.27	COD	2159.4276
SALE21	04-06-2024	CUST1396	PROD43	Electronics	STORE18	Guerrero-Schmitt	1	589.56	0.22	Credit Card	459.8568
SALE22	12-04-2024	CUST1476	PROD69	Clothing	STORE10	Hill, Wagner And Ballan	3	979.42	0.03	PayPal	2850.1122
SALE23	23-06-2024	CUST1368	PROD12	Beauty	STORE8	Bean, Jones And Park	1	719.28	0.08	Credit Card	661.7376
SALE24	08-09-2025	CUST1186	PROD67	Electronics	STORE6	Benson-Elliott	3	77.6	0.07	Credit Card	216.504
SALE25	17-09-2024	CUST1381	PROD5	Clothing	STORE3	Johnson-Wright	4	852.71	0.14	PayPal	2933.3224

Ready Accessibility: Investigate

Step 8: descriptive analysis using Toolpak

AutoSave OFF Cleaned_Ecommerce_Sales_Dataset (Recover... Saved to this PC) Search Upgrade your plan

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design Query

Clipboard Font Alignment Number Styles Cells Editing Add-ins

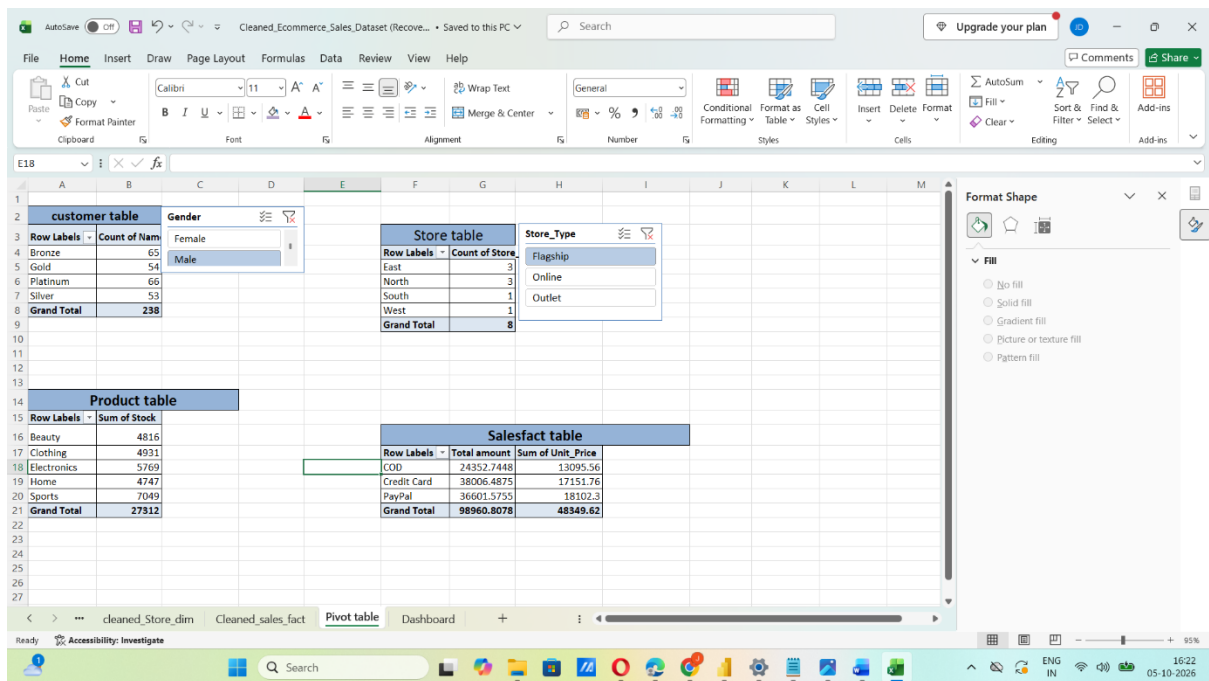
B17 10-12-2024

Store_ID	Store_name	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
STORE3	Johnson-Wright	1	218.04	0.09	Credit Card	198.4164
STORE17	Holloway Ltd	3	104.47	0.27	PayPal	228.7893
STORE8	Bean, Jones And Park	4	914.05	0.25	Credit Card	2742.15
STORE8	Bean, Jones And Park	1	921.51	0.16	COD	774.0684
STORE5	Gray Ltd	3	745.3	0.18	Credit Card	1833.438
STORE7	Martinez-Mullins	3	147.01	0.19	COD	357.2343
STORE4	Alexander Group	2	785.31	0.15	PayPal	1335.027
STORE8	Bean, Jones And Park	4	342.39	0.23	PayPal	1054.5612
STORE5	Gray Ltd	3	448.25	0.07	PayPal	1250.6175
STORE0	Rivera-Warner	2	873.17	0.23	PayPal	1344.6818
STORE14	McDonald Inc	1	278.63	0.07	Credit Card	259.1259
STORE1	Martinez, Adkins And Jc	4	286.92	0.17	Credit Card	952.5744
STORE4	Alexander Group	4	644.58	0.01	Credit Card	2552.5368
STORE16	Frederick-Wright	2	316.9	0.3	PayPal	443.66
STORE15	Shelton Plc	4	403.47	0.29	PayPal	1145.8548
STORE9	Moore, Padilla And But	3	871.34	0.04	COD	2509.4592
STORE19	Mann-Austin	2	451.98	0.25	Credit Card	677.97
STORE4	Alexander Group	2	502.69	0.23	Credit Card	774.1426
STORE4	Alexander Group	2	559.23	0.25	COD	838.845
STORE13	Gardner, Humphrey Ani	4	464.18	0.04	COD	1782.4512
STORE17	Holloway Ltd	3	986.04	0.27	COD	2159.4276
STORE18	Guerrero-Schmitt	1	589.56	0.22	Credit Card	459.8568
STORE10	Hill, Wagner And Ballan	3	979.42	0.03	PayPal	2850.1122
STORE8	Bean, Jones And Park	1	719.28	0.08	Credit Card	661.7376
STORE6	Benson-Elliott	3	77.6	0.07	Credit Card	216.504
STORE3	Johnson-Wright	4	852.71	0.14	PayPal	2933.3224
STORE8	Bean, Jones And Park	4	156.16	0.04	Credit Card	599.6544

Total_Amount	
Descriptive analysis	
Mean	1087.197
Standard Error	18.32724
Median	868.7182
Mode	#N/A
Standard Deviation	819.619
Sample Variance	671775.2
Kurtosis	0.045826
Skewness	0.879047
Range	3889.784
Minimum	18.459
Maximum	3908.243
Sum	2174395
Count	2000

Ready Accessibility: Investigate

Step 9: creating Pivot tables for each table. I have added slices as well.



Step 10: creating Dashboard.

